

INTRODUCTION TO PROJECT MANAGEMENT

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Introduction to Project Management

- Introduction,
- Project and Importance of Project Management,
- Contract Management
- Activities Covered by Software Project Management
- Plans, Methods and Methodologies,
- Some ways of categorizing Software Projects,
- Stakeholders
- Setting Objectives
- Business Case
- Project Success and Failure,
- Management and Management Control
- Project Management life cycle
- Traditional versus Modern Project Management Practices.

Introduction

- whether the management of *software* project is really that different from that of other projects?
- Need to Look at some of the key ideas about the **planning, monitoring and control** of software projects.
- All project are about meeting objectives.
- Like any other project a software project must satisfy all needs.
- To do this we must identify the project's stakeholders and their objectives.
- Ensuring that their objectives are met is the aim of project management.
- However we cannot know that project will meet its objectives in the future unless we know the present of the project.

Why is Software Project Management Important?

- First, A lot of money is at stake with ICT projects.
- In the United Kingdom during the financial year 2002-2003, the central government spent more on contracts for ICT projects than on contracts related to roads (about £2.3 billion as opposed to £1.4 billion).
- The biggest departmental spender was the Department for Work and Pensions, who spent over £800 million on ICT.
- Mismanagement of ICT projects means that there is less to spend on good things such as hospitals.
- Unfortunately, projects are not always successful.
- In a report published in 2003, the Standish Group in the United States analyzed 13,522 projects and concluded that only a third of projects were successful; 82% of projects were late and 43% exceeded their budget.
- The reason for these project shortcomings is often the management of projects.
- Factors causing project failure 'lack of skills and proven approach to project management and risk management'.

1.3 What is a project?

Some dictionary definitions:

- ▣ *“A specific plan or design”*
- ▣ *“A planned undertaking”*
- ▣ *“A large undertaking e.g. a public works scheme”*

Key points above are **planning** and **size** of task

Jobs versus projects

- ❑ **‘Jobs’** – repetition of very well-defined and well understood tasks with very little uncertainty
- ❑ **‘Exploration’** – e.g. finding a cure for cancer: the outcome is very uncertain
- ❑ **‘Projects’** – in the middle!

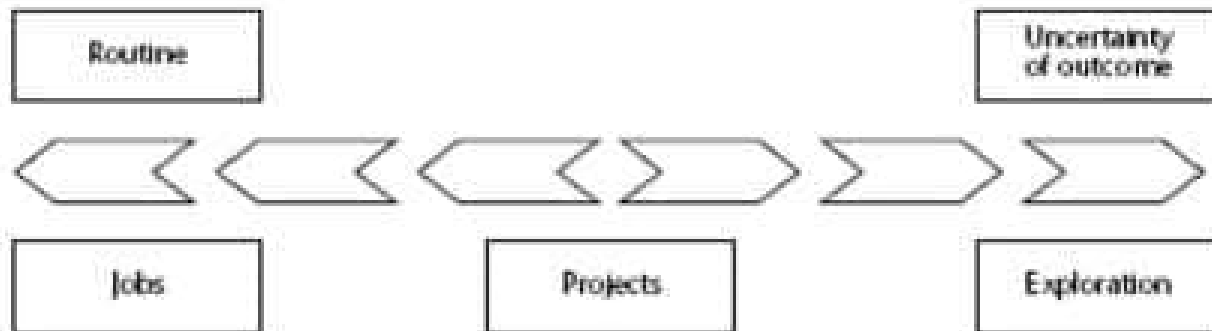


Figure 1.1 Activities most likely to benefit from project management

Jobs Vs Projects

- There is a hazy boundary between the non-routine project and the routine job.
- The first time you do a routine task, it will be like a project.
- On the other hand, a project to develop a system similar to previous ones that you have developed will have a large element of the routine.

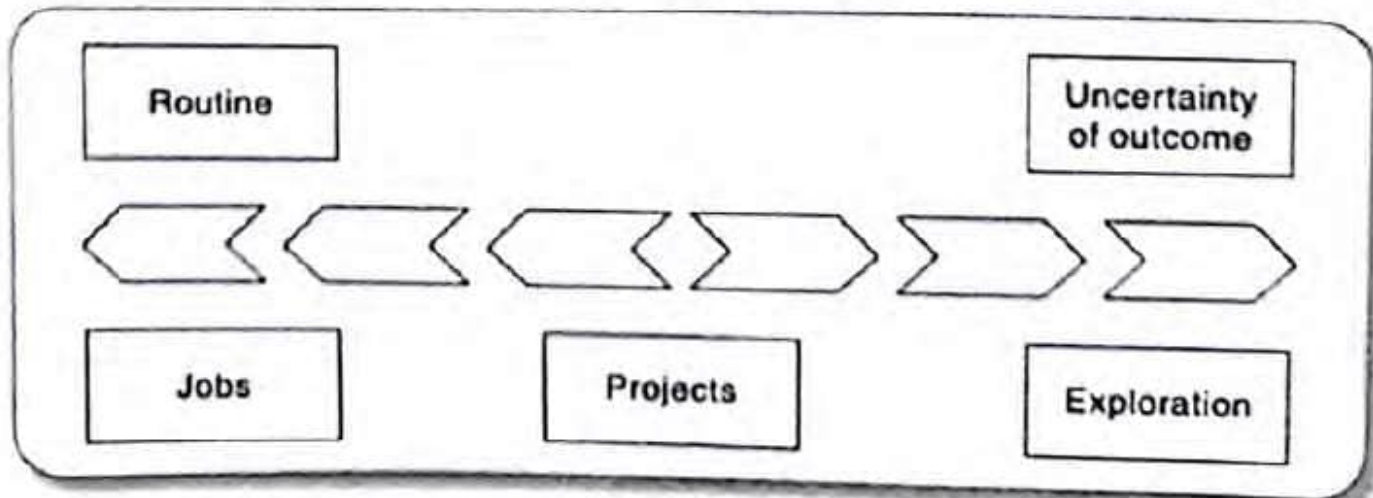


FIGURE 1.1 Activities most likely to benefit from project management

Characteristics of projects

- Non-routine tasks are involved
- Planning is required
- Specific objectives are to be met or a specific product is to be created
- The project has a predetermined time span
- Work is carried out for someone other than yourself
- Work involves several specialisms
- People are formed into a temporary work group to carry out the task
- Work is carried out in several phases
- The resources that are available for use on the project are constrained
- The project is large or complex

1.4 Software Projects versus Other Types of Projects

Fred Brooks identified some characteristics of software projects which make them particularly difficult:

1. Invisibility
2. Complexity
3. Conformity
4. Flexibility

1.5 Contract Management and Technical Project Management

- In-house projects are where the users and the developers of new software work for the same organization.
- However, increasingly organizations contract out ICT development to outside developers.
- Here the **client organization** will often appoint a 'project manager' to supervise the contract who will delegate many technically oriented decisions to the contractors. Thus, the project manager will not worry about estimating the effort needed to write individual software components as long as the overall project is thin budget and on time.
- On the **supplier side** there will need to be project managers who deal with the more technical issue

1.6 Activities Covered by Software Project Management

- Usually there are three successive processes that bring a new system into being
1. **The feasibility study:** requirements, cost and benefits
 2. **Planning:** Stage wise planning
 3. **Project execution:** design and implementation

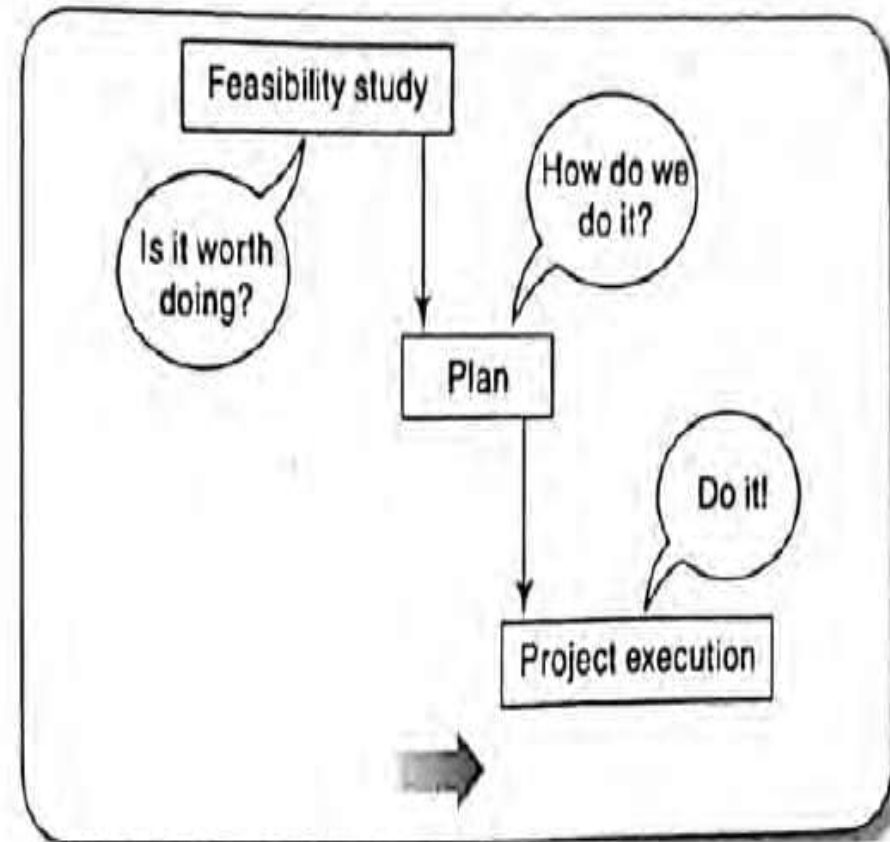
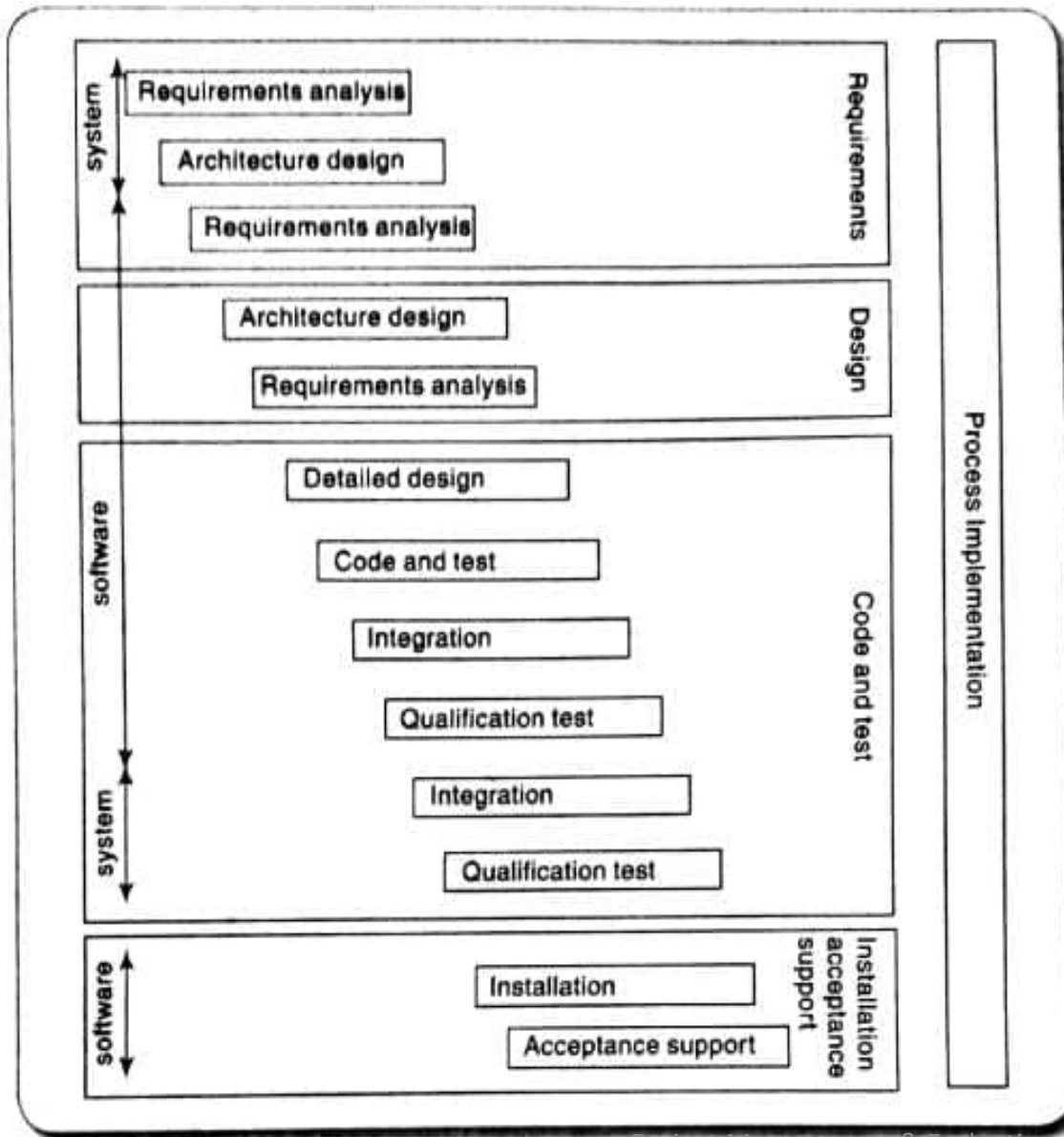


FIGURE 1.2 The feasibility study/plan/execution cycle

1.6 Activities Covered by Software Project Management (Conti..)



- Figure 1.3 shows the typical sequence of software development activities recommended in the international standard ISO 12207.
- Some activities are concerned with the *system* while others relate to *software*.
- The development of software will be only one part of a project

FIGURE 1.3 The ISO 12207 software development life cycle

ISO 12207 life-cycle

1. Requirements analysis

- ▣ Requirements elicitation: what does the client need?
- ▣ Analysis: converting 'customer-facing' requirements into equivalents that developers can understand
- ▣ Requirements will cover
 - Functions, Quality, Resource constraints i.e. costs

2. Architecture design

- ▣ Based on *system requirements*
- ▣ Defines components of system: hardware, software, organizational
- ▣ *Software requirements* will come out of this

3. Detailed Design

4. Code and test

- ▣ Of individual components

5. Integration

- ▣ Putting the components together

ISO 12207 ...

- 6 ☐ Qualification testing
 - ☐ Testing the *system* (not just the *software*)
- 7 ☐ Installation
 - ☐ The process of making the system operational
 - ☐ Includes setting up standing data, setting system parameters, installing on operational hardware platforms, user training etc
- 8 ☐ Acceptance support
 - ☐ Including maintenance and enhancement

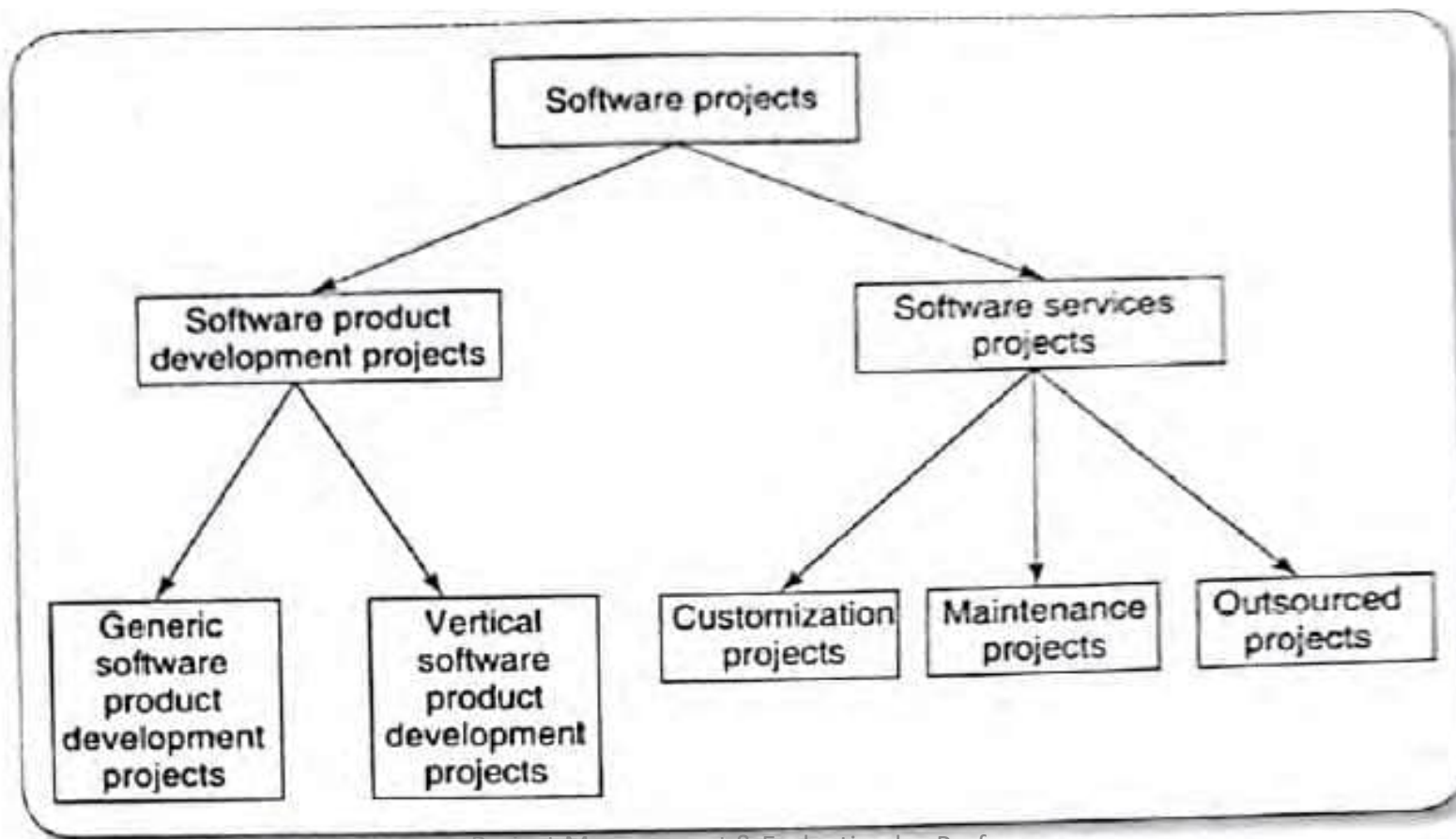
1.7 Plans, Methods and Methodology

- A plan for activity must be based on some idea of a method of work. For example if you were asked to test some software, you may know nothing about the software to be tested, but you could assume that you would need to:
 1. Analyze the requirements for the software.
 2. **Devise** and **write** test cases that will check that each requirement has been satisfied.
 3. Create **test scripts** and **expected results** for each test case.
 4. Compare the actual results and the expected results and identify discrepancies.
- While a method relates to a type of activity in general, a plan takes that method (and perhaps others) and converts it to real activities, identifying for each activity:
 1. *Its* start and end dates
 2. Who will carry it out
 3. What tools and materials - including information - will be needed
- The output from one method might be the input to another. Groups of methods or techniques are often grouped into methodologies such as object-oriented design.

1.8 Some ways of Categorizing Software Projects

- we need to identify the characteristics of a project which could affect the way in which it should be planned and managed. Other factors are discussed below
1. **Changes to the characteristics of software projects:**
 - Reuse of code, customizing and extending existing code, linking library routines and support for framework
 - Customer participation
 2. **Compulsory versus voluntary users:**
 - What the game will do thus depend much on the informed ingenuity of the developers, along with techniques such as market surveys, focus groups and prototype evaluation.
 3. **Information systems versus embedded system**
 - Eg: Stock Control system vs Process control
 4. **Outsourced projects**
 - Lack of expertise, cost-effectively by others
 - Impact on profitability
 5. **Objective driven development**
 - Project may be distinguished by whether aim is to produce a product or to meet certain objective

6. Software products versus services:
- Software Product development projects and software service projects.
 - Ex for Domain specific software products : BANCS from TCS and FINACLE from Infosys
 - Ex for Services: customisation, outsourcing, maintenance, testing and consultancy



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FIGURE 1.4 A classification of software projects

1.10 Stakeholders

- These are people who have a stake or interest in the project
- Their early identification is important as you need to set up adequate communication channels with them.
- Stakeholders can be categorized as:
 1. *internal to the project team* :This means that they will be under the direct managerial control of the project leader.
 2. *External to the project team but within the same organization*: For example, the project leader might need the assistance of the users to carry out systems testing. Here the commitment of the people involved has to be negotiated.
 3. *External to both the project team and the organisation*: External stakeholders may be customers (or users) who will benefit from the system that the project implements. They may be contractors who will *carry out* work for the project. The relationship here is usually based on a contract.
- Different types of stakeholders may have different objectives and one of the jobs of the project leader is to recognize these different interests and to be able to reconcile them.
- *Communication plan* to be created at the start of a project.

1.11 Setting Objectives

- The objectives should define what the project team must achieve for project success.
- Informally the objectives could be written as a set of Statements following the opening words-*'the project will be a success if....'*
- There is often more than one way to meet an objective and the more possible routes to success the better.
- In such a case, a project authority needs to be explicitly identified with overall authority over the project.
- This authority is often a project steering committee (or *project board* or *project management board*) with overall responsibility for setting, monitoring and modifying objectives.
- The project manager runs the project on a day-to-day basis. but regularly reports *to the steering committee*.

Sub-objectives and goals

- **An** effective objective for an individual must be something that is within the control of that individual.
- A more appropriate *goal* or sub-objective for the software developers would be to keep development costs within a certain budget
- We can say that in order to achieve the objective we must achieve certain goals or sub-Objectives first
- **The** mnemonic **SMART** is sometimes used to describe well-defined objectives:

S – specific, that is, concrete and well-defined

M – measurable, that is, satisfaction of the objective can be objectively judged

A – achievable, that is, it is within the power of the individual or group concerned to meet the target

R – relevant, the objective must relevant to the true purpose of the project

T – time constrained: there is defined point in time by which the objective should be achieved

Measure of Effectiveness

- Provides a practical methods of checking that objective has been met.
- Ex: MTBF used to measure reliability
- Performance measurement
- Predictive measure

Ex: large no of errors found during code inspection might indicate potential problem with reliability later.

1.12 The Business Case

- Most projects need to have a justification or business case: the effort and expense of pushing the project through must be seen to be worth while in terms of the benefits that will eventually-be felt.
- A cost-benefit analysis will often be part of the **project's feasibility study**. This will itemize and quantify the project's costs and benefits.
- The benefits will be affected by the completion date: the sooner the project is completed the sooner the benefits can be experienced.
- The quantification of benefits will often require the formulation of a **business model** which explains how the new application can generate the claimed benefits. EX:

Any project plan must ensure that the business case is kept intact. For example:

1. That development costs are not allowed to rise to a level which threatens to exceed the value of benefits
2. That the features of the system are not reduced to a level where the expected benefits cannot be realized
3. That the delivery date is not delayed so that there is an unacceptable loss of benefits

1.13 Project Success and Failure

- The project plan should be designed to ensure project success by preserving the business case for the project.
- Broadly speaking, we can distinguish between *project objectives* and *business objectives*.
- In the case of software projects, project objectives can usually be summarized as delivering:
 1. The agreed functionality
 2. The required level of quality
 3. On time
 4. Within budget
- A project could meet these targets but the application, once delivered could fail to meet the business case.
- On the other hand a project could be late and over budget, but its deliverables could still, over time, generate benefits that outweigh the initial expenditure.

1.13 Project Success and Failure

- Some argue that **the possible gap between project** and business concerns can be reduced by having a broader view of projects that includes business issues.
- Because the focus of project management is, not unnaturally, **on the immediate project**, It may *not* be seen that the project is actually one of a sequence.
- Astute managers may assess which areas of technical expertise it would be beneficial to develop
- **Customer relationships** can also be built up over a number of projects.

1.14 What is management?

This involves the following activities:

- ❑ Planning – deciding what is to be done
- ❑ Organizing – making arrangements
- ❑ Staffing – selecting the right people for the job
- ❑ Directing – giving instructions
- ❑ Monitoring – checking on progress
- ❑ Controlling – taking action to remedy hold-ups
- ❑ Innovating – coming up with solutions when problems emerge
- ❑ Representing – liaising with clients, users, developers and other stakeholders

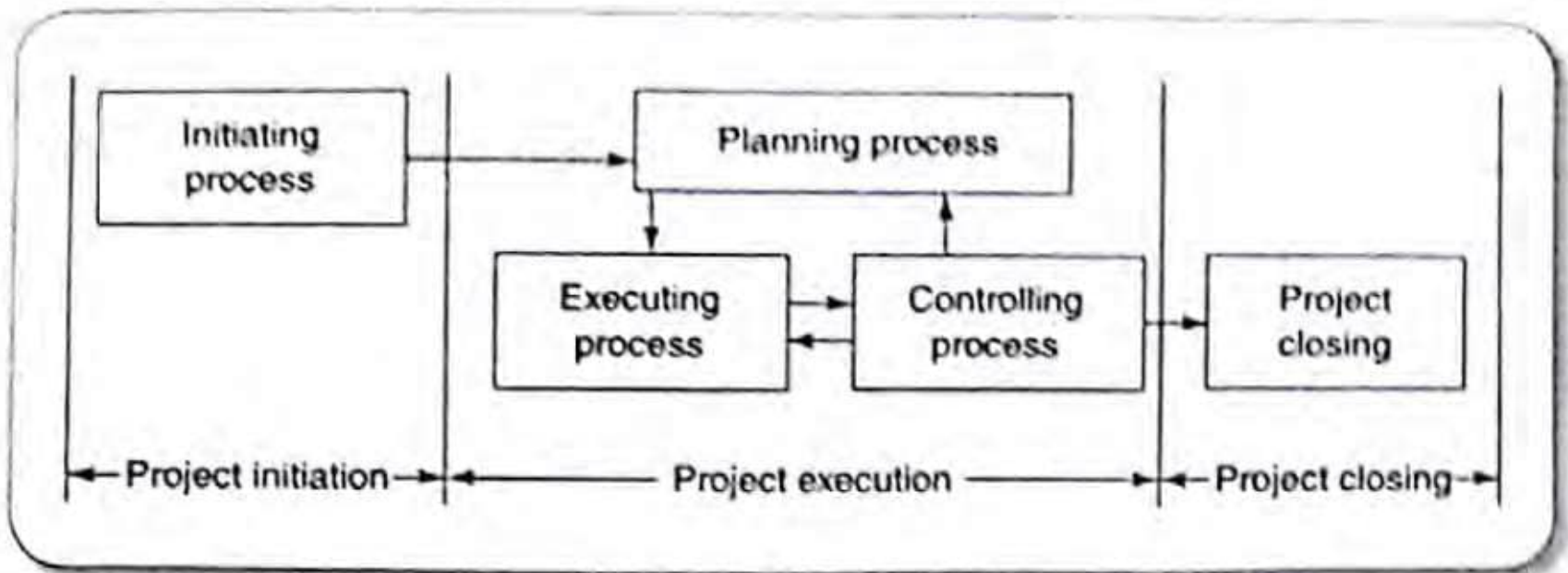


FIGURE 1.5 Principal project management processes

- Figure 1.5 shows this project initiation period.
- Initial project planning involves estimating several characteristics of a Project .
- Based on these estimation , all subsequent project activities are planned.
- The initial project plans are revised periodically as the project progresses and more project data becomes available.
- Once the project execution starts, monitoring and control activities are **taken** up to ensure that the project execution proceeds as planned
- The monitoring activity involves monitoring the progress of the project.
- Control activities are initiated to minimize any significant variation in the plan.

- During project planning. the project manager needs to perform a few well-defined activities that have been outlined below

1. *Estitmaion* : The following project attributes are estimated.

- *Cost* How much is it going to cost to complete the project?
- *Duration* How long is it going to take to complete the project?
- *Effort* How much effort would be necessary for completing the project?

The effectiveness of all activities such as scheduling and staffing. which are planned at a later stage, depends on the accuracy with which the above project parameters have been estimated.

2. *Scheduling*: Based on estimations effort and duration, the schedules for manpower and other resources are developed.

3. *Staffing* : Staff organization and staffing plans are made.

4. *Risk Management*: This activity includes risk identification, analysis and abatement planning.

5. *Miscellaneous Plans* This includes making several other plans such as **quality assurance plan. configuration management plan**. etc.

1.15 Management Control

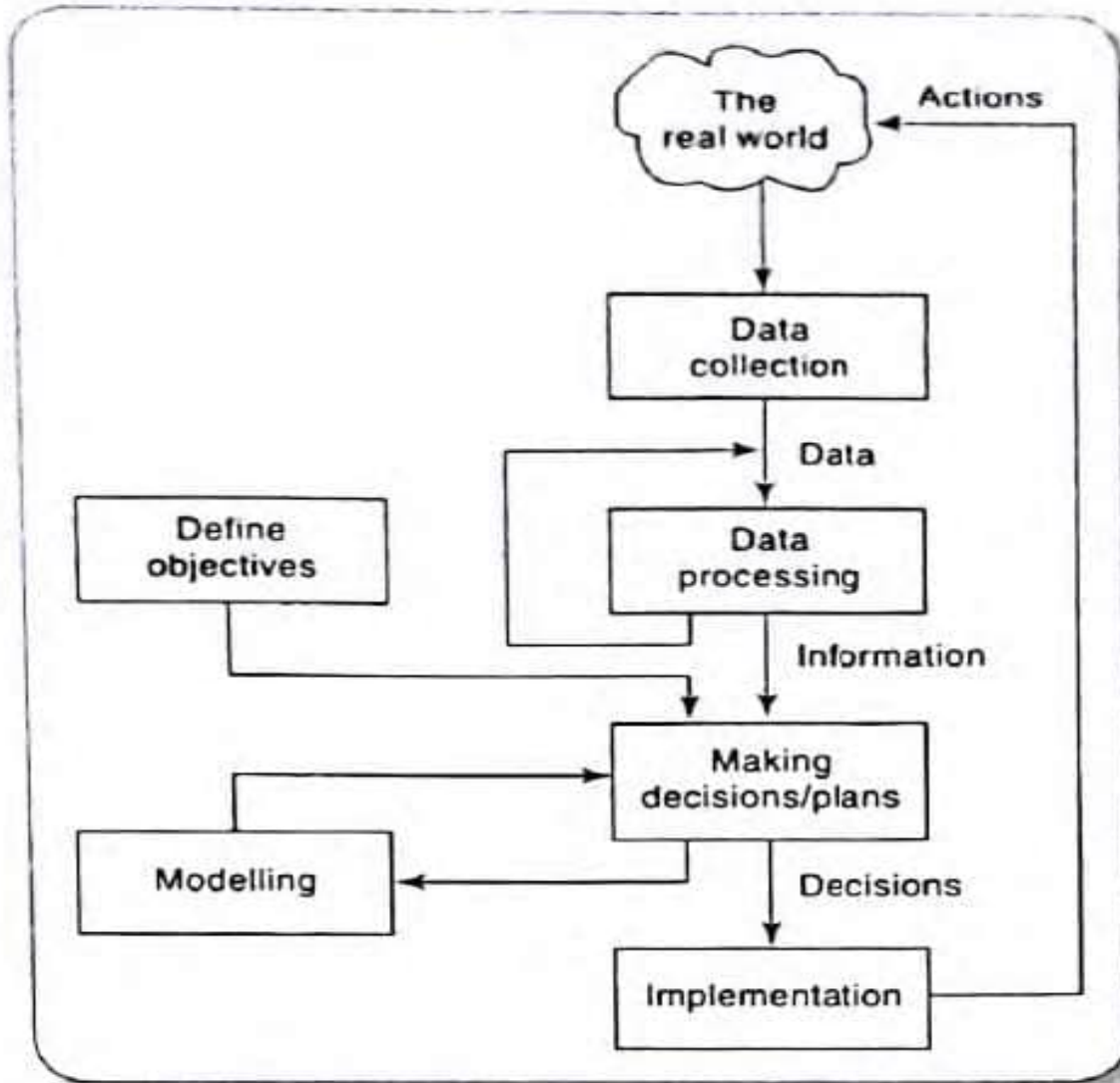


FIGURE 1.6 The project control cycle

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Management control

- ❑ **Data – the raw details**
e.g. *'6,000 documents processed at location X'*
- ❑ **Information – the data is processed to produce something that is meaningful and useful**
e.g. *'productivity is 100 documents a day'*
- ❑ **Comparison with objectives/goals**
e.g. *'we will not meet target of processing all documents by 31st March'*
- ❑ **Modelling – working out the probable outcomes of various decisions**
e.g. *'if we employ two more staff at location X how quickly can we get the documents processed?'*
- ❑ **Implementation – carrying out the remedial actions that have been decided upon**

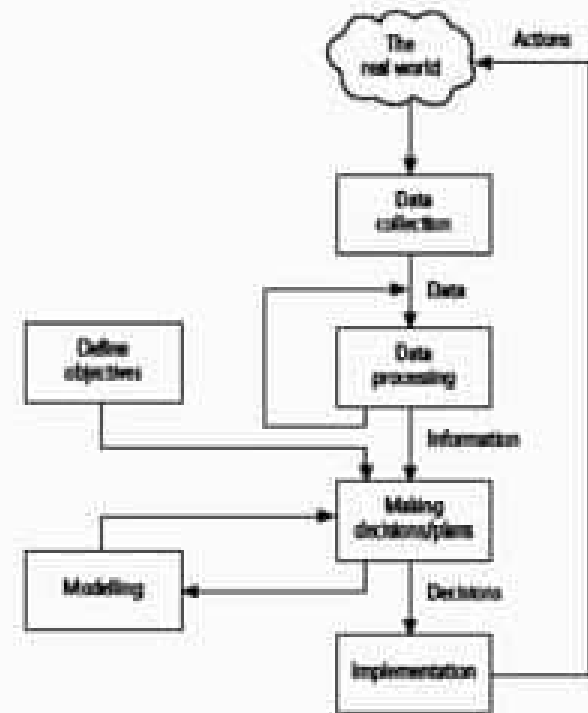
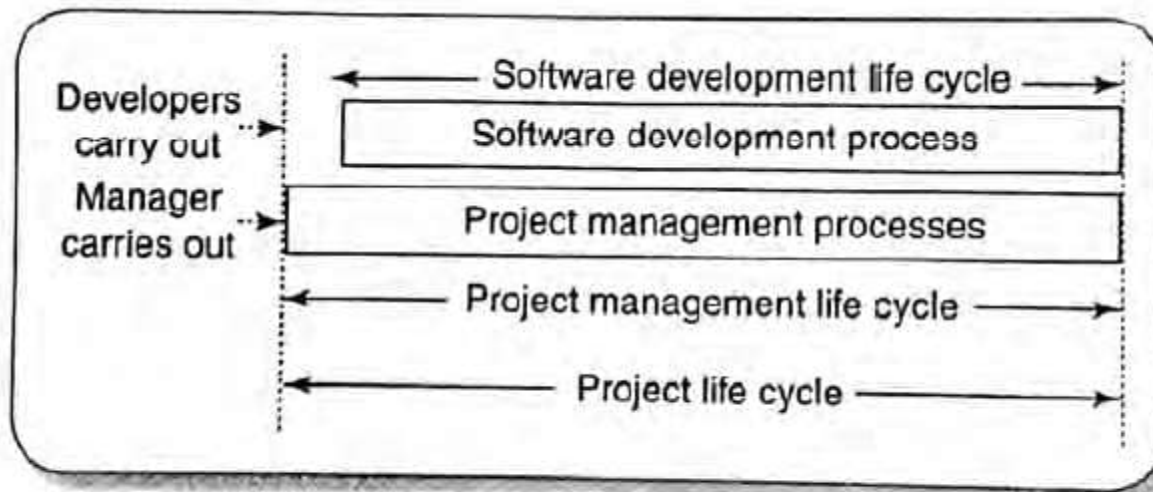
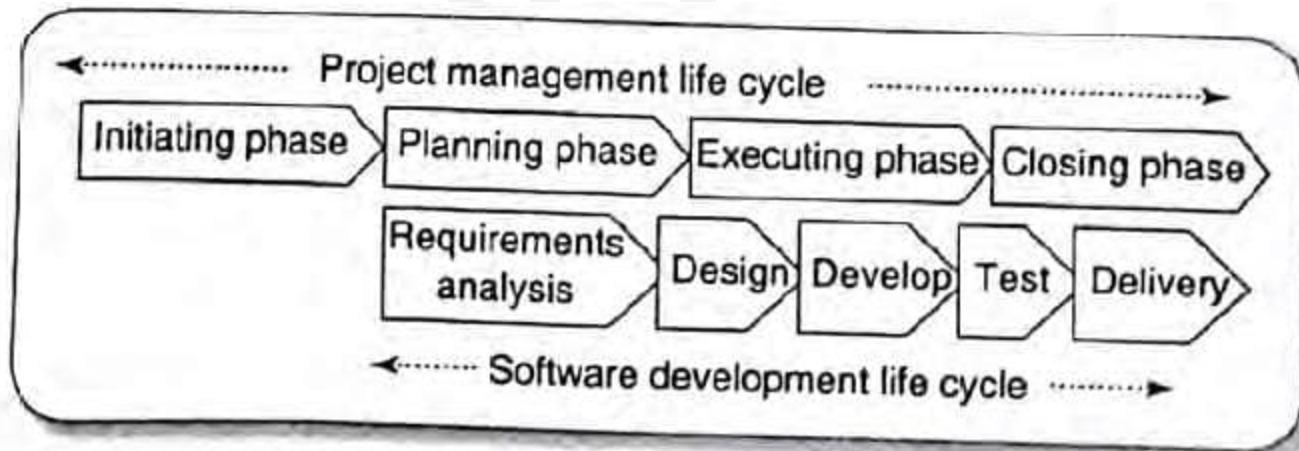


Figure 1.4 The project control cycle.

Software Development and Project Management life cycles



Project management life cycle versus software development life cycle



Different phases of project management life cycle and software development life cycle

1.16 Project Management Life Cycle

- The different phases of the project management life cycle are shown in Figure 1.8. In the following, **we** discuss the main activities that are carried out in each phase.

1. Project Initiation:

- The project initiation phase usually starts with project concept development
- The different aspects of the project that are investigated and understood include: **the scope of the project, project constraints, the cost that would be incurred and the benefits that would accrue.**
- Based on this understanding, a **feasibility study** is undertaken to determine whether the project would be financially and technically feasible.
- For any new project based on the **feasibility study**, the **business case** is developed. Once the top management agrees to the business case, the **project manager is appointed, the project charter** is written, and finally **project team** is formed
- Barry Boehm summarized the questions that need to be asked and answered in order to have an understanding of these project characteristics.

Project Initiation (Conti..)

W5HH Principle: This set of seven questions is the following:

- Why is the software being built?
- What will be done?
- When will it be done?
- Who is responsible for a function?
- Where are they organizationally located?
- How will the job be done technically and managerially?
- How much of each resource is needed?

Project Initiation (Conti..)

Project Bidding: Once an organization's top management is convinced by the business case, the project charter is developed. For some categories of projects, it may be necessary to have a formal bidding process to select a suitable vendor based on some cost-performance criteria

- i. *Re quest for quotation (RFQ) :*** *SOW(Statement of work), RFT(Request For Tender)*
- ii. *Request for proposal (RFP) :***
 - Many times it so happens that an organization has reasonable understanding of the problem to be solved, however it does not have a good grasp of the solution aspects.
 - In this case, the organization may solicit solution proposals from vendors. The vendors may submit a few alternative solutions and the approximate costs for each solution.
 - Based on the RFP process, the requesting organization can form a clear idea of the project solutions required, based on which it can form a statement work (SOW) for requesting RFQ from the vendors.
- iii. *Request for Information(RFI) :***
 - An organization soliciting bids may publish an RFI.
 - it must be noted that vendor selection is seldom done based on RFI, but the RFI response from the vendors may be used in conjunction with RFP and RFQ responses for vendor selection.

1.16 Project Management Life Cycle(Conti..)

2. Project Planning

An important outcome of the project initiation phase is the project charter.

During the project planning phase, the project manager comes out several processes and creates the following documents:

1. *Project plans*: This document identifies the **project tasks, and a schedule for the project tasks** that assigns project resources and time frames to the tasks.
2. *Resource plan*: It lists the resources, manpower and equipment that would be required to execute the project
3. *Financial plan*: It documents the plan for manpower, equipment and other costs.
4. *Quality plan*: Plan of **quality targets** and **control plans** are included in this document.
5. *Risk plan*: This document lists the identification of the potential risks, their prioritization and a plan for the actions that would be taken to contain the different risks.

1.16 Project Management Life Cycle(Conti..)

3. Project execution

- A series of management processes are undertaken
- Monitoring and control processes are executed
- Project plan may be **revised periodically**
- Quality of the deliverables is ensured through execution of proper processes
- Project execution phase completes and the project closure phase starts

1.16 Project Management Life Cycle(Conti..)

4. Project Closure

- Involves completing the release of all the required deliverables to the customer along with the necessary documentation
- Subsequently all the project resources are released and supply agreements with the vendors are terminated and all the pending payments are completed
- Finally a post implementation review is undertaken .

1.17 Traditional versus Modern Project Management Practices

- Over the last two decades, the basic approach taken by the software industry to develop software has undergone a radical change.
 - Software development projects are increasingly being based on either tailoring some existing product or reusing certain pre built libraries
 - In either case, two important goals are i) maximize code reuse ii) Compression of project duration
 - In the following section we will discuss some important differences between project management practices and traditional practices
1. Planning Incremental Delivery
 2. Quality Management
 3. Change Management
 4. Requirements Management
 5. Release Management
 6. Risk Management
 7. Scope Management

Project Evaluation

1. Introduction
2. Evaluation of Individual Projects
3. Cost Benefit Evaluation Techniques
4. Risk Evaluation

2.1 Introduction

- The first that many developers hear of an ICT project is when they are allocated to the project team.
- However, new projects do not appear out of thin air. There will be some process - varying in sophistication between organizations - that decides that the project is worth doing.
- Sometimes a project to establish an ICT infrastructure within an organization might not deliver a direct financial benefit, but could provide a platform for subsequent projects to do so.
- This chapter will discuss some ways in which projects can be evaluated and compared for inclusion in a project portfolio.

2.4 Evaluation of Individual Projects

- We will now look more closely at how the feasibility of an individual project can be evaluated.
 1. Technical assessment:
 2. Cost benefit Analysis
 3. Cash flow forecasting

2.4 Evaluation of Individual Projects(Conti..)

1. Technical Assessment:

- Technical assessment of a proposed system consists of evaluating whether the required functionality can be achieved with current affordable technologies.
- The costs of the technology adopted must be taken into account in the cost-benefit analysis.

2.4 Evaluation of Individual Projects(Conti..)

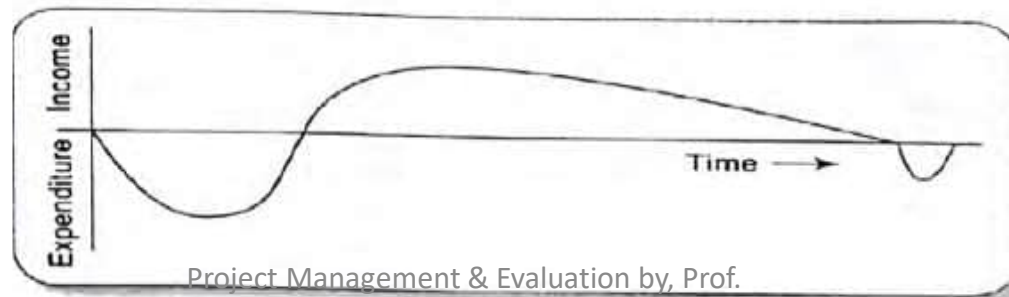
2. Cost-benefit analysis

- Even where the estimated benefits will exceed the estimated costs, it is often necessary to decide if the proposed project is the best of several options.
- Cost-benefit analysis comprises two steps:
 1. Identifying all of the costs and benefits of carrying out the project and operating the delivered application These include the development costs, the operating costs, and the benefits expected from the new system. Where the proposed system is a replacement, these estimates should reflect the change in costs and benefits due to the new system
 2. Expressing these costs and benefits in common units .We must express each cost and benefit - and the net benefit which is the difference between the two-in money.
- Most direct costs are easy to quantify in monetary terms and can be categorized as:
 1. *Development costs*, including development staff costs.
 2. *Setup costs*, consisting of the costs of putting the system into place, mainly of any new hardware but also including the costs of file conversion, recruitment and staff training.
 3. *Operational casts* relating to operating the system after installation.

2.4 Evaluation of Individual Projects(Conti..)

3. Cash Flow Forecasting

- As important as estimating the overall costs and benefits of a project is producing a cash flow forecast which indicates when expenditure and income will take place(Figure 2.1).
- We need to spend money, such as staff wages, during a project's development.
- We need to know that we can fund this development expenditure either from the company's own resources or by borrowing .
- A forecast is needed of when expenditure, such as the payment of salaries, and any income are to be expected.
- Accurate cash flow forecasting is difficult, as it is done early in the project's lifecycle and many items to be estimated might be some years in the future.
- When estimating future cash flows, it is usual to ignore the effects of inflation.



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FIGURE 2.1 Typical product life cycle cash flow

2.5 Cost – benefit Evaluation Techniques

- We now take a look at some methods for comparing projects on the basis of their cash flow forecasts.
- Table 2.1 illustrates cash flow forecasts for four projects.
- In each case, it is assumed that the cash flows take place at the end of each year.
- For short-term projects or where there are significant seasonal cash flow patterns, quarterly, or even monthly, cash flow forecasts could be appropriate.

TABLE 2.1 Four project cash flow projections – figures are end of year totals (£)

Year	Project 1	Project 2	Project 3	Project 4
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	75,000
Net profit	50,000	100,000	50,000	75,000

2.5 Cost – benefit Evaluation Techniques (Conti..)

1. Net Profit:

- The net profit of a project is the difference between the total costs and the total income over the life of the
- Moreover, the simple net profit takes no account of the timing of the cash
- P1 and P3 have equal net profit.
- Having to wait for a return has the disadvantage that the investment must be funded for longer.
- Add to that the fact that, other things being equal, estimates in the more distant future are less reliable than short-term estimates and we can see that the two projects are not equally preferable.

TABLE 2.1 Four project cash flow projections – figures are end of year totals (£)

Year	Project 1	Project 2	Project 3	Project 4
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	75,000
Net profit	50,000	100,000	50,000	75,000

2.5 Cost – benefit Evaluation Techniques (Conti..)

2. Payback Period:

- The payback period is the time taken to break even or pay back the initial investment.
- Normally, the project with the shortest payback period will be chosen on the basis that an organization will wish to minimize the time that a project is 'in debt'
- The advantage of the payback period is that it is simple to calculate and is not particularly sensitive to small forecasting errors.
- Its disadvantage as a selection technique is that it ignores the overall profitability of the project-in fact, it totally ignores any income (or expenditure) once the project has broken even.
- Thus the fact that projects 2 and 4 are, overall, more profitable than project 3 is ignored.

2.5 Cost – benefit Evaluation Techniques (Conti..)

3. Return on investment

- The return on investment (ROI), also known as the accounting rate of return (ARR) provides a way of comparing the net profitability to the investment required.
- There are some variations on the formula used to calculate the return on investment but a straightforward common version is:
$$\text{ROI} = (\text{Average annual profit} / \text{Total investment}) \times 100$$
- The return on investment provides a simple, easy-to-calculate measure of return on capital.
- two severe disadvantages.
 - Like the net profitability, it takes no account of the timing of the cash flows.
 - More importantly, this rate of return bears no relationship to the interest rates offered or charged by banks (or any other normal interest rate) since it takes no account of the timing of the cash flows or of the compounding of interest. It is therefore, potentially, very misleading.

2.5 Cost – benefit Evaluation Techniques (Conti..)

4. Net Present Value

- The calculation of net present value is a project evaluation technique that takes into account the profitability of a project and the timing of the cash flows that are produced.
- This is based on the view that receiving £100 today is better than having to wait until next year to receive it.
- The annual rate by which we discount future earnings is known as the discount rate-10%.
- The present value of any future cash flow may be obtained by applying the following formula

$$\text{Present value} = \frac{\text{value in year } t}{(1 + r)^t}$$

where r is the discount rate, expressed as a decimal value, and t is the number of years into the future that the cash flow occurs.

2.5 Cost – benefit Evaluation Techniques (Conti..)

4. Net Present Value (Conti..)

- Alternatively, and rather more easily, the present value of a cash flow may be calculated by multiplying the cash flow by the appropriate discount factor.
- The NPV for a project is obtained by discounting each cash flow (both negative and positive) and summing the discounted values. It is normally assumed that a initial investment takes place immediately (indicated as year 0 and is not discounted) Later cash flows are normally assumed to take place at the end of each year and are discounted by the appropriate amount.
- The main difficulty with NPV for deciding between projects is selecting an appropriate discount rate.
- The exact discount rate is normally less important than ensuring that the same discount rate is used for all projects being compared.

2.5 Cost – benefit Evaluation Techniques (Conti..)

4. Net Present Value (Conti..)

TABLE 2.2 NPV discount factors

Year	Discount rate (%)					
	5	6	8	10	12	15
1	0.9524	0.9434	0.9259	0.9091	0.8929	0.8696
2	0.9070	0.8900	0.8573	0.8264	0.7972	0.7561
3	0.8638	0.8396	0.7938	0.7513	0.7118	0.6575
4	0.8227	0.7921	0.7350	0.6830	0.6355	0.5718
5	0.7835	0.7473	0.6806	0.6209	0.5674	0.4972
6	0.7462	0.7050	0.6302	0.5645	0.5066	0.4323
7	0.7107	0.6651	0.5835	0.5132	0.4523	0.3759
8	0.6768	0.6274	0.5403	0.4665	0.4039	0.3269
9	0.6446	0.5919	0.5002	0.4241	0.3606	0.2843
10	0.6139	0.5584	0.4632	0.3855	0.3220	0.2472
15	0.4810	0.4173	0.3152	0.2394	0.1827	0.1229
20	0.3769	0.3118	0.2145	0.1486	0.1037	0.0611
25	0.2953	0.2330	0.1460	0.0923	0.0588	0.0304

2.5 Cost – benefit Evaluation Techniques (Conti..)

4. Net Present Value (Conti..)

TABLE 2.3 Applying the discount factors to project 1

Year	Project 1 cash flow (£)		Discount factor @ 10%	Discounted cash flow (£)	
0	-100,000	- 100,000	1.0000	-100,000	-100 000
1	10,000	30 000	0.9091	9,091	27 273
2	10,000	30 000	0.8264	8,264	24 792
3	10,000	30 000	0.7513	7,513	22 539
4	20,000	30 000	0.6830	13,660	20 490
5	100,000	30 000	0.6209	62,090	18 627
Net Profit	£50,000		NPV: £618		13 721

4. Net Present Value (Conti..)

- The main difficulty with NPV for deciding between projects is selecting an appropriate discount rate.
- Some organizations have a standard rate but, where this is not the case, then the discount rate should be chosen to reflect available interest .
- The exact discount rate is normally less important than ensuring that the same discount rate is used for all projects being compared.
- However, it is important to check that the ranking of projects is not sensitive to small changes in the discount rate

2.5 Cost – benefit Evaluation Techniques (Conti..)

5. Internal rate of return

- NVP might not be directly comparable with earnings from other investments or the costs of borrowing capital, Such costs are usually quoted as a percentage interest rate.
- The internal rate of return (IRR) attempts to provide a profitability measure as a percentage return that is directly comparable with interest rates.
- The IRR is calculated as that percentage discount rate that would produce an NPV of zero.
- It is most easily calculated using a spreadsheet or other computer program that provides functions for calculating the IRR.
- One deficiency of the IRR is that it does not indicate the absolute size of the return.
- Another objection to the internal rate of return is that, under certain conditions, it is possible to find more than one rate that will produce a zero NPV

2.5 Cost – benefit Evaluation Techniques (Conti..)

5. Internal rate of return(Continued)

- NPV and IRR are not, however, a complete answer to economic project evaluation.
1. A total evaluation must also take into account **the problems of funding the cash flows** - will we, for example, be able to repay the interest on any borrowed money at the appropriate time?
 2. While a project's IRR might indicate a profitable project, future earnings from a relatively risky project might be far less reliable than earnings from, say, investing with a bank. We might **undertake a more detailed risk analysis** .
 3. **We must also consider any one project** within the financial and economic framework of the organization as a whole - if we fund this one, will we also be able to fund other worthy projects?

2.6 Risk Evaluation

- Project Risk and Business Risks

1. Risk Identification and Ranking:

- In any project evaluation we should identify the risks and quantify their effects.
- One approach is to construct project risk matrix utilizing a checklist of possible risks and classifying risks according to their relative importance and likelihood.
- Table 2.5 illustrates a basic project risk matrix listing some of the business risks for a project, with their importance and likelihood classified as high (H), medium (M), low (L) or exceedingly unlikely (-)
- So that projects may be compared, the list of risks must be the same for each project assessed.

TABLE 2.5 A fragment of a basic project/business risk matrix for an e-commerce application

Risk	Importance	Likelihood
Client rejects proposed look and feel of site	H	—
Competitors undercut prices	H	M
Warehouse unable to deal with increased demand	M	L
Online payment has security problems	M	M
Maintenance costs higher than estimated	L	L
Response times deter purchasers	M	M

2.6 Risk Evaluation(Conti..)

2. Risk and net present value

- Where a project is relatively risky, it is a common practice to use a higher discount rate to calculate net present value.
- This risk premium might, for example, be an additional 2% for a reasonably safe project or 5% for a fairly risky one.
- Projects may be categorized as high, medium, or low risk using a **scoring method and premiums designated** for each category.
- The premiums, even if arbitrary, provide a consistent method of taking risk into account.

2.6 Risk Evaluation(Conti..)

3. Cost – benefit analysis

- A rather more sophisticated approach to the evaluation of risk is to consider each possible outcome and estimate the probability of its occurring and the corresponding value of the outcome.
- Rather than a cash flow forecast for a project, we will then have a set of cash flow forecasts, each with an associated probability of occurring.
- The value of the project is then obtained by summing the cost or benefit for each possible outcome weighted by its corresponding probability
- This approach is frequently used to evaluate large projects such as the building of motorways(Uncertain)
- this approach is more appropriate for the evaluation of a portfolio of projects where overall profitability is the primary concern, more successful projects can offset the impact of less successful ones.

3. Cost – benefit analysis(Conti..)

Exercise 2.8 BuyRight, a software house, is considering developing a payroll application for use in academic institutions and is currently engaged in a cost-benefit analysis. Study of the market has shown that, if BuyRight can target it efficiently and no competing products become available, it will obtain a high level of sales generating an annual income of £800,000. It estimates that there is a 1 in 10 chances of this happening. However, a competitor might launch a competing application before its own launch date and then sales might generate only £100,000 per year. It estimates that there is a 30% chance of this happening. The most likely outcome, it believes, is somewhere in between these two extremes – it will gain a market lead by launching before any competing product becomes available and achieve an annual income of £650,000. Buy Right has therefore calculated its expected sales income as in Table 2.6.

TABLE 2.6 BuyRight's income forecasts

Sales	Annual sales income (£)	Probability	Expected value (£)
	i	p	$i \times p$
High	800,000	0.1	80,000
Medium	650,000	0.6	390,000
Low	100,000	0.3	30,000
Expected Income			500,000

Development costs are estimated at £750,000. Sales levels are expected to be constant for at least four years. Annual costs of marketing and product maintenance are estimated at £200,000, irrespective of the market share. Would you advise going ahead with the project?

2.6 Risk Evaluation(Conti..)

4. Risk Profile Analysis

- An approach which attempts to overcome some of the objections to cost-benefit averaging is the construction of risk profiles using sensitivity analysis.
- This involves varying each of the **parameters** that affect the project's cost or benefits to ascertain how sensitive the project's profitability is to each factor.
- We might, for example, vary one of our original estimates by plus or minus 5% and recalculate the expected costs and benefits for the project.
- By repeating this exercise for each of our estimates in turn we can evaluate the sensitivity of the project to each factor.
- By studying the results of a sensitivity analysis we can identify those factors that are most important to the success of the project.
- We then need to decide whether we can exercise greater control over them or otherwise mitigate their effects.
- If neither is the case, then we must live with the risk or abandon the project.

2.6 Risk Evaluation(Conti..)

5. Using decision trees

- There are many situations, however, where we can evaluate whether a risk is important and, if it is, decide a suitable course of action.
- at any point, it is important to be able to assess how a decision will affect the future profitability of the project.
- As an example, say a successful company is considering when to replace its sales order processing system.
 - The decision largely rests upon the rate at which its business expands - if its market share significantly increases the existing system might need to be replaced within two years. Not replacing the system in time could be an expensive option as it could lead to lost revenue if it cannot cope with increased sales. Replacing the system immediately will, however, be expensive as it will mean deferring other projects already scheduled. a
 - It is calculated that extending the existing system will have an NPV of £75,000, although if the market expands significantly, this will be turned into a loss with an NPV of -£1 00,000 due to lost revenue.
 - If the market does expand, replacing the system now has an NPV of £250,000 due to the benefits of being able to handle increased sales and other benefits such as improved management information. If sales do not increase, however, the benefits will be severely reduced and the project will suffer a loss with an NPV of -£50,000.
 - The company estimate the likelihood of the market increasing significantly at 20% and, hence, the probability that it will not increase at 80%. This scenario can be represented as a tree structure as shown in next figure.

5. Using decision trees (Conti..)

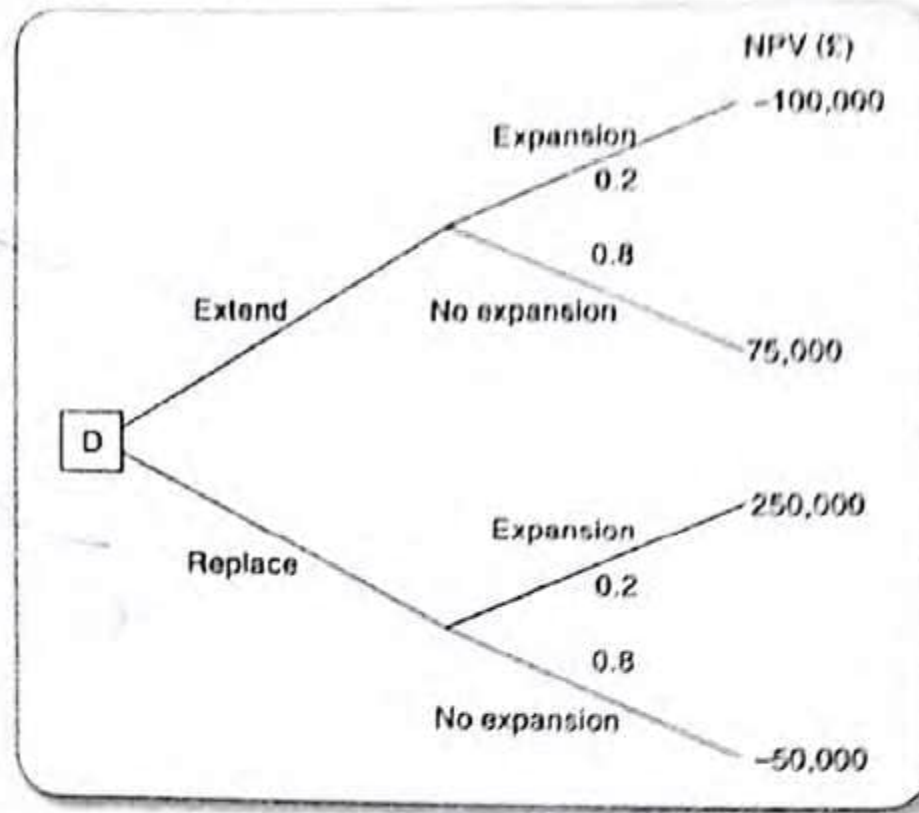


FIGURE 2.2 A decision tree

- The expected value of each path is the sum of the value of each possible outcome multiplied by its probability of occurrence.
- The expected value of extending the system is therefore £40,000 i.e. $(75,000 \times 0.8 - 100,000 \times 0.2)$
- and the expected value of replacing the system £10,000 $(250,000 \times 0.2 - 50,000 \times 0.8)$.
- The company should therefore choose the option of extending the existing system.